Maple Code Documentation for the Key Vanishing Lemma (Curve Complement)

Logarithmic Case

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Introduction

This document provides a detailed, line-by-line explanation of the Maple code used in the paper *Improved Degree Bounds for Hyperbolicity of Surfaces and Curve Complements* by Lei Hou, Dinh Tuan Huynh, Joël Merker, and Song-Yan Xie (arXiv:2603.14881). Specifically, this code implements the computation for the logarithmic case (curve complement) of the Key Vanishing Lemma. The code systematically verifies the non-existence of nonzero negatively twisted invariant 2-jet differentials of order $m = 8$ on the complement of a curve of degree 13 in $\mathbb{P}^2$ (the code variable `d` is set to 12, so the curve degree is $\mathtt{d} + 1 = 13$), subject to a specified twist parameter $\mathsf{Twist} = -7$. The computation involves solving algebraic systems to enforce the vanishing of the differential to high order along the divisor at infinity. The code presented here is the full working version; the reader can copy and execute each block sequentially in a Maple worksheet to reproduce the results.

The parameters m, `d`, Twist, and the equation shape (via the auxiliary polynomial SS) can be freely adjusted to test other cases. The iteration up to $y^{22\cdot\mathtt{d}}$ is necessary because the Key Vanishing Lemma for the logarithmic case requires verifying divisibility up to order $2m - 1 = 15$ in the notation of the paper; however, the algebraic system becomes tractable only after expanding to a sufficiently high order. In practice, we iterate until the number of unknown coefficients stabilizes to constants (all `nops` outputs become 1), which for the chosen parameters occurs at $22 \cdot \mathtt{d}$. This high iteration count is also a preparation for future computations with $d = 11$ (where the required order would be even larger); the current code structure allows such experiments by simply modifying the initialization block.

Part 1: Initialization and Parameter Definition

Explanation: Set the basic parameters. The command `restart` clears the Maple session. The variable $m = 8$ is the order of the jet differential. The twist parameter is set to -7. The variable `d` is set to 12; this means the degree of the curve in $\mathbb{P}^2$ is $\mathtt{d} + 1 = 13$. The variable `degree` is defined as `d`+1 but is not directly used in the computation.

```
restart:
m := 8;
d := 12;
degree := d + 1:
twist := -7:
```

Listing 1: Initialization code

Part 2: Defining the Auxiliary Polynomial and the Curve Equation

Explanation: Define the auxiliary polynomial SS. In the logarithmic case, the computation is performed on a fibration where the base is the curve. The polynomial SS is chosen as:

$$SS := x^{13} + x^6 + 1$$

The equation defining the curve in the affine chart $\mathbb{C}^2$ (with coordinates x, y) is:

$$RR = y^{13} + SS = 0$$

The variable x is the coordinate on the curve, and y is the fiber coordinate.

```
SS := x^(d+1) + x^6 + 1:

RR := y^(d+1) + SS;
```

Listing 2: Auxiliary polynomial and curve equation

Part 3: Template for the Jet Differential

Explanation: Construct the local expression of an invariant jet differential Jet_xR on the affine chart. It consists of sums over k (the power of the Wronskian Δ_{xR}) and j (the power of R_1/RR, which corresponds to the derivative of the fiber coordinate). The coefficients $AA_j, BB_j, \ldots, HH_j$ are unknown polynomials in x and y (later expanded) that will be determined. The factors $1/Ry^{m-k\cdot 2}$ ensure the appropriate homogeneity. Here $m = 8$, and we compute:

- $k = 0$: $m - 0 \cdot 2 = 8$, $m - 0 \cdot 3 = 8$
- $k = 1$: $m - 1 \cdot 2 = 6$, $m - 1 \cdot 3 = 5$
- $k = 2$: $m - 2 \cdot 2 = 4$, $m - 2 \cdot 3 = 2$
- $k = 3$: $m - 3 \cdot 2 = 2$, $m - 3 \cdot 3 = -1$
- $k = 4$: $m - 4 \cdot 2 = 0$, $m - 4 \cdot 3 = -4$
- $k = 5$: $m - 5 \cdot 2 = -2$, $m - 5 \cdot 3 = -7$
- $k = 6$: $m - 6 \cdot 2 = -4$, $m - 6 \cdot 3 = -10$
- $k = 7$: $m - 7 \cdot 2 = -6$, $m - 7 \cdot 3 = -13$

The sums range over $j = 0$ to $m - 3k$. Since $m - 3k$ becomes negative for $k \geq 3$, those sums are empty.

```
Jet_xR := expand(
    + (1/Ry^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (R1/RR)^j * Delta_xR^0, j=0..m-0*3)
    + (1/Ry^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (R1/RR)^j * Delta_xR^1, j=0..m-1*3)
    + (1/Ry^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (R1/RR)^j * Delta_xR^2, j=0..m-2*3)
    + (1/Ry^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (R1/RR)^j * Delta_xR^3, j=0..m-3*3)
    + (1/Ry^(m-4*2)) * add(EE||j * x1^(m-4*3-j) * (R1/RR)^j * Delta_xR^4, j=0..m-4*3)
    + (1/Ry^(m-5*2)) * add(FF||j * x1^(m-5*3-j) * (R1/RR)^j * Delta_xR^5, j=0..m-5*3)
    + (1/Ry^(m-6*2)) * add(GG||j * x1^(m-6*3-j) * (R1/RR)^j * Delta_xR^6, j=0..m-6*3)
    + (1/Ry^(m-7*2)) * add(HH||j * x1^(m-7*3-j) * (R1/RR)^j * Delta_xR^7, j=0..m-7*3)
);
```

Listing 3: Jet expression construction

Part 4: Degree Bounds for Unknown Polynomials

Explanation: Compute the maximal allowed degrees in y for each unknown polynomial coefficient. The formulas are:

$$dA[l] = (m - 0 \cdot 2) \cdot \mathtt{d} - (m - 0 \cdot 3 - l) \cdot 2 - l \cdot 1 - 0 \cdot 4 + \mathsf{Twist}$$

and similarly for the other blocks. Here l is the index of the coefficient (the power of x_1). These bounds determine the maximum degree of the unknown polynomials in y and later in x.

```
for l from 0 to m-0*3 do dA[l] := (m-0*2)*d - (m-0*3-l)*2 - l*1 - 0*4 + twist: od:
for l from 0 to m-1*3 do dB[l] := (m-1*2)*d - (m-1*3-l)*2 - l*1 - 1*4 + twist: od:
for l from 0 to m-2*3 do dC[l] := (m-2*2)*d - (m-2*3-l)*2 - l*1 - 2*4 + twist: od:
for l from 0 to m-3*3 do dD[l] := (m-3*2)*d - (m-3*3-l)*2 - l*1 - 3*4 + twist: od:
for l from 0 to m-4*3 do dE[l] := (m-4*2)*d - (m-4*3-l)*2 - l*1 - 4*4 + twist: od:
for l from 0 to m-5*3 do dF[l] := (m-5*2)*d - (m-5*3-l)*2 - l*1 - 5*4 + twist: od:
for l from 0 to m-6*3 do dG[l] := (m-6*2)*d - (m-6*3-l)*2 - l*1 - 6*4 + twist: od:
for l from 0 to m-7*3 do dH[l] := (m-7*2)*d - (m-7*3-l)*2 - l*1 - 7*4 + twist: od:
```

Listing 4: Degree bounds for each block

Part 5: Wronskian and Transition Formulas

Explanation: Define the Wronskian $\Delta_{xR} = x_1 \cdot (R_2/RR - R_1 \cdot R_1/RR^2) - x_2 \cdot (R_1/RR)$. The formulas $Solx1$ and $Solx2$ express x_1 and x_2 in terms of y_1, R_1, R_2 and partial derivatives of RR. These come from the implicit function theorem applied to the curve equation $RR = 0$.

```
Delta_xR := x1 * (R2/RR - R1*R1/RR^2) - x2 * (R1/RR):

Solx1 := - y1*Ry/Rx + R1/Rx:

Solx2 := - Rxx*y1^2*Ry^2/Rx^3 + 2*Rxx*y1*Ry*R1/Rx^3 - Rxx*R1^2/Rx^3
        + 2*y1^2*Rxy*Ry/Rx^2 - 2*y1^2*Ry/Rx - y2*Ry/Rx + R2/Rx:
```

Listing 5: Wronskian and transition formulas

Part 6: Transition to the Other Chart

Explanation: Substitute the expressions for x_1 and x_2 into Jet_xR to express it in terms of R_1, y_1, Δ_{Ry} (the Wronskian in the alternative chart). The command `mtaylor` performs a multivariate Taylor expansion up to order 100, which is far beyond the needed degree. The result is sorted by the variables.

```
Jet_Ry := sort(mtaylor(
    expand(subs(R2 = - (RR^2*Delta_Ry - RR*R1*y2 - Jet_xR), Jet_xR)),
    [R1, y1, Delta_Ry], 100), [R1, y1, Delta_Ry]):
```

Listing 6: Transition to the other chart

Part 7: Extract Coefficients from Taylor Expansion

Explanation: Extract coefficients of specific monomials $R_1^{m-j-3k} y_1^j \Delta_{Ry}^k$ from the Taylor expansion. The factor $Ry^{m-j-3k} \cdot RR^{m-j-3k} \cdot Rx^{m-2k}$ is multiplied to clear denominators, converting rational expressions into polynomial equations. The result is stored in $EEq[i, j, k]$.

```
printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEq[m-j-3*k,j,k] := factor(
            Ry^(m-j-3*k) * RR^(m-j-3*k) * Rx^(m-2*k) *
            coeftayl(Jet_Ry, [R1,y1,Delta_Ry]=[0,0,0], [m-j-3*k,j,k])
        ):
    od:
od:
```

Listing 7: Coefficient extraction

Part 8: Define the Curve Equation and Partial Derivatives

Explanation: Define the specific curve RR in $\mathbb{C}^2$ (affine chart) and compute its partial derivatives. The equation is:

$$RR = y^{13} + x^{13} + x^6 + 1 = 0$$

All first and second partial derivatives are computed symbolically for later use.

```
RR := y^(d+1) + SS;

Rx := diff(RR,x): Ry := diff(RR,y):

Rxx := diff(RR,x,x): Rxy := diff(RR,x,y): Ryy := diff(RR,y,y):
```

Listing 8: Curve definition and derivatives

Part 9: Expand Equations as Power Series in y

Explanation: Expand each EEq as a power series in y, truncating at order $(m-j-3k)\cdot\mathtt{d}$. This corresponds to the expected vanishing order along the divisor at infinity. The variables are recorded.

```
printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], y, (m-j-3*k)*d):
        Var[m-j-3*k,j,k] := indets(EEq[m-j-3*k,j,k]) minus {x,y}:
    od:
od:
```

Listing 9: Power series expansion in y

Part 10: Initialize Unknown Polynomials (First Expansion)

Explanation: Initialize the unknown polynomials $AA_j, BB_j, \ldots, HH_j$ as polynomials in x and y with undetermined coefficients. The expansion is truncated at degree $1 \cdot d - 1$ in y. This is the first of many expansions; subsequent iterations will increase the degree bound.

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],1*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],1*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],1*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],1*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],1*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],1*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],1*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],1*d-1)): od:
```

Listing 10: Initialize unknown polynomials (first step)

Part 11: First Iteration — Divisibility by $y^{1\cdot\mathtt{d}}$

Explanation: In the first iteration, we require that the equations $Eq[1, m-1-3k, k]$ be divisible by $y^{1\cdot\mathtt{d}} = y^{12}$. We collect all coefficients (in x, y) from the Taylor expansion in y up to order $\mathtt{d} - 1 = 11$. These

coefficients form a linear system in the unknown constants. The system is solved, and the solutions are assigned back to the unknowns.

```
Systemey1d := {}:
for k from 0 to (m-1)/3 do
    Coeffs_y1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,k], y, 1*d)), [x,y])}:
    Systemey1d := Systemey1d union Coeffs_y1d[1,m-1-3*k,k]:
od:
Systeme_y1d := Systemey1d:

Variables_y1d := indets(Systeme_y1d):
Solutions_y1d := solve(Systeme_y1d, Variables_y1d):
assign(Solutions_y1d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 11: First iteration: solving divisibility by $y^{1\cdot \mathtt{d}}$

Part 12: Iterations for $k = 2$ to $k = 22$

Explanation: The code proceeds iteratively, increasing the order of divisibility by $y^{k\cdot \mathtt{d}}$ for $k = 2, 3, \ldots, 22$. At each iteration, the unknown polynomials are re-expanded to a higher degree in y (up to $k \cdot \mathtt{d} - 1$), and the divisibility condition is enforced by collecting coefficients from the Taylor expansion and solving the resulting linear system. The solutions from the previous iteration are already substituted. The output after each iteration shows the number of terms remaining in each coefficient polynomial, gradually reducing until all become 1.

The iterations for $k = 2$ through $k = 21$ are structurally identical; only the order parameter changes. For completeness, we list each iteration explicitly. The final iteration at $k = 22$ is also shown; it yields all `nops` outputs equal to 1, indicating that all undetermined constants are forced to zero.

Iteration $k = 2$: Divisibility by $y^{2\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],2*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],2*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],2*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],2*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],2*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],2*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],2*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],2*d-1)): od:

Systemey2d := {}:
for k from 0 to (m-2)/3 do
    Coeffs_y2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,k], y, 2*d)), [x,y])}:
    Systemey2d := Systemey2d union Coeffs_y2d[2,m-2-3*k,k]:
od:
Systeme_y2d := Systemey2d:
```

```
Variables_y2d := indets(Systeme_y2d):
Solutions_y2d := solve(Systeme_y2d, Variables_y2d):
assign(Solutions_y2d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 12: Iteration 2: solving divisibility by $y^{2\cdot \mathtt{d}}$

Iteration $k = 3$: Divisibility by $y^{3\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],3*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],3*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],3*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],3*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],3*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],3*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],3*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],3*d-1)): od:

Systemey3d := {}:
for k from 0 to (m-3)/3 do
    Coeffs_y3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,k], y, 3*d)), [x,y])}:
    Systemey3d := Systemey3d union Coeffs_y3d[3,m-3-3*k,k]:
od:
Systeme_y3d := Systemey3d:

Variables_y3d := indets(Systeme_y3d):
Solutions_y3d := solve(Systeme_y3d, Variables_y3d):
assign(Solutions_y3d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 13: Iteration 3: solving divisibility by $y^{3\cdot \mathtt{d}}$

Iteration $k = 4$: Divisibility by $y^{4\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],4*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],4*d-1)): od:
```

```
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],4*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],4*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],4*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],4*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],4*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],4*d-1)): od:

Systemey4d := {}:
for k from 0 to (m-4)/3 do
    Coeffs_y4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,k], y, 4*d)), [x,y])}:
    Systemey4d := Systemey4d union Coeffs_y4d[4,m-4-3*k,k]:
od:
Systeme_y4d := Systemey4d:

Variables_y4d := indets(Systeme_y4d):
Solutions_y4d := solve(Systeme_y4d, Variables_y4d):
assign(Solutions_y4d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 14: Iteration 4: solving divisibility by $y^{4\cdot \mathtt{d}}$

Iteration $k = 5$: Divisibility by $y^{5\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],5*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],5*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],5*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],5*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],5*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],5*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],5*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],5*d-1)): od:

Systemey5d := {}:
for k from 0 to (m-5)/3 do
    Coeffs_y5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,k], y, 5*d)), [x,y])}:
    Systemey5d := Systemey5d union Coeffs_y5d[5,m-5-3*k,k]:
od:
Systeme_y5d := Systemey5d:

Variables_y5d := indets(Systeme_y5d):
Solutions_y5d := solve(Systeme_y5d, Variables_y5d):
assign(Solutions_y5d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
```

```
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 15: Iteration 5: solving divisibility by $y^{5\cdot \mathrm{d}}$

Iteration $k = 6$: Divisibility by $y^{6\cdot \mathrm{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],6*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],6*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],6*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],6*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],6*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],6*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],6*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],6*d-1)): od:

Systemey6d := {}:
for k from 0 to (m-6)/3 do
    Coeffs_y6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,k], y, 6*d)), [x,y])}:
    Systemey6d := Systemey6d union Coeffs_y6d[6,m-6-3*k,k]:
od:
Systeme_y6d := Systemey6d:

Variables_y6d := indets(Systeme_y6d):
Solutions_y6d := solve(Systeme_y6d, Variables_y6d):
assign(Solutions_y6d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 16: Iteration 6: solving divisibility by $y^{6\cdot \mathrm{d}}$

Iteration $k = 7$: Divisibility by $y^{7\cdot \mathrm{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],7*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],7*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],7*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],7*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],7*d-1)): od:
```

```
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],7*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],7*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],7*d-1)): od:

Systemey7d := {}:
for k from 0 to (m-7)/3 do
    Coeffs_y7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,k], y, 7*d)), [x,y])}:
    Systemey7d := Systemey7d union Coeffs_y7d[7,m-7-3*k,k]:
od:
Systeme_y7d := Systemey7d:

Variables_y7d := indets(Systeme_y7d):
Solutions_y7d := solve(Systeme_y7d, Variables_y7d):
assign(Solutions_y7d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 17: Iteration 7: solving divisibility by $y^{7\cdot \mathtt{d}}$

Iteration $k = 8$: Divisibility by $y^{8\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],8*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],8*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],8*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],8*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],8*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],8*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],8*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],8*d-1)): od:

Systemey8d := {}:
for k from 0 to (m-8)/3 do
    Coeffs_y8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,k], y, 8*d)), [x,y])}:
    Systemey8d := Systemey8d union Coeffs_y8d[8,m-8-3*k,k]:
od:
Systeme_y8d := Systemey8d:

Variables_y8d := indets(Systeme_y8d):
Solutions_y8d := solve(Systeme_y8d, Variables_y8d):
assign(Solutions_y8d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
```

```
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 18: Iteration 8: solving divisibility by $y^{8\cdot \mathtt{d}}$

Iteration $k = 9$: Divisibility by $y^{9\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],9*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],9*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],9*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],9*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],9*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],9*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],9*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],9*d-1)): od:

Systemey9d := {}:
for k from 0 to (m-9)/3 do
    Coeffs_y9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,k], y, 9*d)), [x,y])}:
    Systemey9d := Systemey9d union Coeffs_y9d[9,m-9-3*k,k]:
od:
Systeme_y9d := Systemey9d:

Variables_y9d := indets(Systeme_y9d):
Solutions_y9d := solve(Systeme_y9d, Variables_y9d):
assign(Solutions_y9d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 19: Iteration 9: solving divisibility by $y^{9\cdot \mathtt{d}}$

Iteration $k = 10$: Divisibility by $y^{10\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],10*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],10*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],10*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],10*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],10*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],10*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],10*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],10*d-1)): od:
```

```
Systemey10d := {}:
for k from 0 to (m-10)/3 do
    Coeffs_y10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10-3*k,k], y, 10*d)),
        [x,y])}:
    Systemey10d := Systemey10d union Coeffs_y10d[10,m-10-3*k,k]:
od:
Systeme_y10d := Systemey10d:

Variables_y10d := indets(Systeme_y10d):
Solutions_y10d := solve(Systeme_y10d, Variables_y10d):
assign(Solutions_y10d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 20: Iteration 10: solving divisibility by $y^{10\cdot \mathrm{d}}$

Iteration $k = 11$: Divisibility by $y^{11\cdot \mathrm{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],11*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],11*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],11*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],11*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],11*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],11*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],11*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],11*d-1)): od:

Systemey11d := {}:
for k from 0 to (m-11)/3 do
    Coeffs_y11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11-3*k,k], y, 11*d)),
        [x,y])}:
    Systemey11d := Systemey11d union Coeffs_y11d[11,m-11-3*k,k]:
od:
Systeme_y11d := Systemey11d:

Variables_y11d := indets(Systeme_y11d):
Solutions_y11d := solve(Systeme_y11d, Variables_y11d):
assign(Solutions_y11d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 21: Iteration 11: solving divisibility by $y^{11\cdot \mathrm{d}}$

Iteration $k = 12$: Divisibility by $y^{12\cdot d}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],12*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],12*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],12*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],12*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],12*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],12*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],12*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],12*d-1)): od:

Systemey12d := {}:
for k from 0 to (m-12)/3 do
    Coeffs_y12d[12,m-12-3*k,k] := {coeffs(expand(mtaylor(Eq[12,m-12-3*k,k], y, 12*d)),
        [x,y])}:
    Systemey12d := Systemey12d union Coeffs_y12d[12,m-12-3*k,k]:
od:
Systeme_y12d := Systemey12d:

Variables_y12d := indets(Systeme_y12d):
Solutions_y12d := solve(Systeme_y12d, Variables_y12d):
assign(Solutions_y12d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 22: Iteration 12: solving divisibility by $y^{12\cdot d}$

Iteration $k = 13$: Divisibility by $y^{13\cdot d}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],13*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],13*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],13*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],13*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],13*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],13*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],13*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],13*d-1)): od:

Systemey13d := {}:
for k from 0 to (m-13)/3 do
    Coeffs_y13d[13,m-13-3*k,k] := {coeffs(expand(mtaylor(Eq[13,m-13-3*k,k], y, 13*d)),
        [x,y])}:
    Systemey13d := Systemey13d union Coeffs_y13d[13,m-13-3*k,k]:
```

```
od:
Systeme_y13d := Systemey13d:

Variables_y13d := indets(Systeme_y13d):
Solutions_y13d := solve(Systeme_y13d, Variables_y13d):
assign(Solutions_y13d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 23: Iteration 13: solving divisibility by $y^{13\cdot \mathtt{d}}$

Iteration $k = 14$: Divisibility by $y^{14\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],14*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],14*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],14*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],14*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],14*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],14*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],14*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],14*d-1)): od:

Systemey14d := {}:
for k from 0 to (m-14)/3 do
    Coeffs_y14d[14,m-14-3*k,k] := {coeffs(expand(mtaylor(Eq[14,m-14-3*k,k], y, 14*d)),
        [x,y])}:
    Systemey14d := Systemey14d union Coeffs_y14d[14,m-14-3*k,k]:
od:
Systeme_y14d := Systemey14d:

Variables_y14d := indets(Systeme_y14d):
Solutions_y14d := solve(Systeme_y14d, Variables_y14d):
assign(Solutions_y14d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 24: Iteration 14: solving divisibility by $y^{14\cdot \mathtt{d}}$

Iteration $k = 15$: Divisibility by $y^{15\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],15*d-1)): od:
```

```
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],15*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],15*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],15*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],15*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],15*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],15*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],15*d-1)): od:

Systemey15d := {}:
for k from 0 to (m-15)/3 do
    Coeffs_y15d[15,m-15-3*k,k] := {coeffs(expand(mtaylor(Eq[15,m-15-3*k,k], y, 15*d)),
        [x,y])}:
    Systemey15d := Systemey15d union Coeffs_y15d[15,m-15-3*k,k]:
od:
Systeme_y15d := Systemey15d:

Variables_y15d := indets(Systeme_y15d):
Solutions_y15d := solve(Systeme_y15d, Variables_y15d):
assign(Solutions_y15d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 25: Iteration 15: solving divisibility by $y^{15\cdot \mathrm{d}}$

Iteration $k = 16$: Divisibility by $y^{16\cdot \mathrm{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],16*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],16*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],16*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],16*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],16*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],16*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],16*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],16*d-1)): od:

Systemey16d := {}:
for k from 0 to (m-16)/3 do
    Coeffs_y16d[16,m-16-3*k,k] := {coeffs(expand(mtaylor(Eq[16,m-16-3*k,k], y, 16*d)),
        [x,y])}:
    Systemey16d := Systemey16d union Coeffs_y16d[16,m-16-3*k,k]:
od:
Systeme_y16d := Systemey16d:

Variables_y16d := indets(Systeme_y16d):
```

```
Solutions_y16d := solve(Systeme_y16d, Variables_y16d):
assign(Solutions_y16d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 26: Iteration 16: solving divisibility by $y^{16\cdot \mathtt{d}}$

Iteration $k = 17$: Divisibility by $y^{17\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],17*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],17*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],17*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],17*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],17*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],17*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],17*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],17*d-1)): od:

Systemey17d := {}:
for k from 0 to (m-17)/3 do
    Coeffs_y17d[17,m-17-3*k,k] := {coeffs(expand(mtaylor(Eq[17,m-17-3*k,k], y, 17*d)),
        [x,y])}:
    Systemey17d := Systemey17d union Coeffs_y17d[17,m-17-3*k,k]:
od:
Systeme_y17d := Systemey17d:

Variables_y17d := indets(Systeme_y17d):
Solutions_y17d := solve(Systeme_y17d, Variables_y17d):
assign(Solutions_y17d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 27: Iteration 17: solving divisibility by $y^{17\cdot \mathtt{d}}$

Iteration $k = 18$: Divisibility by $y^{18\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],18*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],18*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],18*d-1)): od:
```

```
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],18*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],18*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],18*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],18*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],18*d-1)): od:

Systemey18d := {}:
for k from 0 to (m-18)/3 do
    Coeffs_y18d[18,m-18-3*k,k] := {coeffs(expand(mtaylor(Eq[18,m-18-3*k,k], y, 18*d)),
        [x,y])}:
    Systemey18d := Systemey18d union Coeffs_y18d[18,m-18-3*k,k]:
od:
Systeme_y18d := Systemey18d:

Variables_y18d := indets(Systeme_y18d):
Solutions_y18d := solve(Systeme_y18d, Variables_y18d):
assign(Solutions_y18d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 28: Iteration 18: solving divisibility by $y^{18\cdot \mathrm{d}}$

Iteration $k = 19$: Divisibility by $y^{19\cdot \mathrm{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],19*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],19*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],19*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],19*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],19*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],19*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],19*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],19*d-1)): od:

Systemey19d := {}:
for k from 0 to (m-19)/3 do
    Coeffs_y19d[19,m-19-3*k,k] := {coeffs(expand(mtaylor(Eq[19,m-19-3*k,k], y, 19*d)),
        [x,y])}:
    Systemey19d := Systemey19d union Coeffs_y19d[19,m-19-3*k,k]:
od:
Systeme_y19d := Systemey19d:

Variables_y19d := indets(Systeme_y19d):
Solutions_y19d := solve(Systeme_y19d, Variables_y19d):
assign(Solutions_y19d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
```

```
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 29: Iteration 19: solving divisibility by $y^{19\cdot \mathtt{d}}$

Iteration $k = 20$: Divisibility by $y^{20\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],20*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],20*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],20*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],20*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],20*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],20*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],20*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],20*d-1)): od:

Systemey20d := {}:
for k from 0 to (m-20)/3 do
    Coeffs_y20d[20,m-20-3*k,k] := {coeffs(expand(mtaylor(Eq[20,m-20-3*k,k], y, 20*d)),
        [x,y])}:
    Systemey20d := Systemey20d union Coeffs_y20d[20,m-20-3*k,k]:
od:
Systeme_y20d := Systemey20d:

Variables_y20d := indets(Systeme_y20d):
Solutions_y20d := solve(Systeme_y20d, Variables_y20d):
assign(Solutions_y20d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 30: Iteration 20: solving divisibility by $y^{20\cdot \mathtt{d}}$

Iteration $k = 21$: Divisibility by $y^{21\cdot \mathtt{d}}$

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],21*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],21*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],21*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],21*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],21*d-1)): od:
```

```
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],21*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],21*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],21*d-1)): od:

Systemey21d := {}:
for k from 0 to (m-21)/3 do
    Coeffs_y21d[21,m-21-3*k,k] := {coeffs(expand(mtaylor(Eq[21,m-21-3*k,k], y, 21*d)),
        [x,y])}:
    Systemey21d := Systemey21d union Coeffs_y21d[21,m-21-3*k,k]:
od:
Systeme_y21d := Systemey21d:

Variables_y21d := indets(Systeme_y21d):
Solutions_y21d := solve(Systeme_y21d, Variables_y21d):
assign(Solutions_y21d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 31: Iteration 21: solving divisibility by $y^{21\cdot \mathtt{d}}$

Iteration $k = 22$: Divisibility by $y^{22\cdot \mathtt{d}}$ (Final)

```
for l from 0 to m-0*3 do AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j),
    j=0..min(dA[l],22*d-1)): od:
for l from 0 to m-1*3 do BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j),
    j=0..min(dB[l],22*d-1)): od:
for l from 0 to m-2*3 do CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j),
    j=0..min(dC[l],22*d-1)): od:
for l from 0 to m-3*3 do DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j),
    j=0..min(dD[l],22*d-1)): od:
for l from 0 to m-4*3 do EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j),
    j=0..min(dE[l],22*d-1)): od:
for l from 0 to m-5*3 do FF||l := add(add(F||l[i,j]*x^i*y^j, i=0..dF[l]-j),
    j=0..min(dF[l],22*d-1)): od:
for l from 0 to m-6*3 do GG||l := add(add(G||l[i,j]*x^i*y^j, i=0..dG[l]-j),
    j=0..min(dG[l],22*d-1)): od:
for l from 0 to m-7*3 do HH||l := add(add(H||l[i,j]*x^i*y^j, i=0..dH[l]-j),
    j=0..min(dH[l],22*d-1)): od:

Systemey22d := {}:
for k from 0 to (m-22)/3 do
    Coeffs_y22d[22,m-22-3*k,k] := {coeffs(expand(mtaylor(Eq[22,m-22-3*k,k], y, 22*d)),
        [x,y])}:
    Systemey22d := Systemey22d union Coeffs_y22d[22,m-22-3*k,k]:
od:
Systeme_y22d := Systemey22d:

Variables_y22d := indets(Systeme_y22d):
Solutions_y22d := solve(Systeme_y22d, Variables_y22d):
assign(Solutions_y22d):

for l from 0 to m-0*3 do nops(expand(AA||l)) od;
for l from 0 to m-1*3 do nops(expand(BB||l)) od;
for l from 0 to m-2*3 do nops(expand(CC||l)) od;
for l from 0 to m-3*3 do nops(expand(DD||l)) od;
for l from 0 to m-4*3 do nops(expand(EE||l)) od;
```

```
for l from 0 to m-5*3 do nops(expand(FF||l)) od;
for l from 0 to m-6*3 do nops(expand(GG||l)) od;
for l from 0 to m-7*3 do nops(expand(HH||l)) od;
```

Listing 32: Iteration 22: solving divisibility by $y^{22\cdot \mathtt{d}}$

Part 13: Final Verification

Explanation: After completing all iterations, the coefficients have been fully determined. The final expression for the jet differential Jet_xR can be reconstructed. Since all unknown constants have been forced to zero, the resulting differential is identically zero, confirming the vanishing result.

```
Jet_xR := factor(
    + (1/Ry^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (R1/RR)^j * Delta_xR^0, j=0..m-0*3)
    + (1/Ry^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (R1/RR)^j * Delta_xR^1, j=0..m-1*3)
    + (1/Ry^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (R1/RR)^j * Delta_xR^2, j=0..m-2*3)
    + (1/Ry^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (R1/RR)^j * Delta_xR^3, j=0..m-3*3)
    + (1/Ry^(m-4*2)) * add(EE||j * x1^(m-4*3-j) * (R1/RR)^j * Delta_xR^4, j=0..m-4*3)
    + (1/Ry^(m-5*2)) * add(FF||j * x1^(m-5*3-j) * (R1/RR)^j * Delta_xR^5, j=0..m-5*3)
    + (1/Ry^(m-6*2)) * add(GG||j * x1^(m-6*3-j) * (R1/RR)^j * Delta_xR^6, j=0..m-6*3)
    + (1/Ry^(m-7*2)) * add(HH||j * x1^(m-7*3-j) * (R1/RR)^j * Delta_xR^7, j=0..m-7*3)
):

Vars := indets(Jet_xR) minus {x, x1, x2, y, y1, y2, R1, R2, RR}:
NbVars := nops(Vars);
```

Listing 33: Final verification

Conclusion

The Maple code systematically constructs an invariant jet differential of order 8 on the complement of a degree 13 curve in $\mathbb{P}^2$ with a twist of -7. The solution is obtained by imposing vanishing conditions order by order in y and solving linear algebraic systems. The final verification shows that all undetermined coefficients are forced to be zero, demonstrating that no nonzero such differential exists for these parameters.